

PENDINE, United KingdomWMO: **036080**Lat: **51.750N**Lon: **4.517W**Elev: **5**StdP: **101.26**Time Zone: **0.00 (EUW)**Period: **88-99**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.4	-1.1	-4.4	2.6	1.1	-3.4	2.8	1.5	13.5	9.7	12.5	9.5	2.0	50	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	23.8	17.8	22.0	16.8	20.4	16.3	18.9	22.0	17.9	20.4	17.1	19.4	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.7	20.1	16.9	12.0	18.9	16.1	11.4	18.0	53.5	21.7	50.3	20.4	47.9	19.3	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.1	8.9	-4.3	25.9	1.3	1.8	-5.3	27.2	-6.0	28.2	-6.8	29.3	-7.8	30.6		
			-4.8	20.1	1.8	1.5	-6.0	21.2	-7.1	22.0	-8.1	22.9	-9.4	24.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.8	5.9	7.3	8.3	11.5	13.5	16.0	15.9	13.6	11.2	8.0	5.5	(6)
(7)		DBStd	4.54	2.80	2.80	2.13	2.33	2.83	2.06	2.06	2.43	1.83	2.82	2.96	3.22	(7)
(8)		HDD10.0	646	132	116	86	59	16	1	0	0	1	21	72	142	(8)
(9)		HDD18.3	2968	390	348	342	301	212	146	76	82	143	222	309	397	(9)
(10)		CDD10.0	733	1	1	2	9	62	107	187	183	107	58	13	3	(10)
(11)		CDD18.3	14	0	0	0	0	1	1	5	8	0	0	0	0	(11)
(12)		CDH23.3	77	0	0	0	0	1	3	27	46	0	0	0	0	(12)
(13)		CDH26.7	9	0	0	0	0	0	0	1	8	0	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.5	4.5	4.2	4.1	3.7	3.9	3.3	3.4	4.1	4.5	3.8	4.2	(14)
(15)	Precipitation	PrecAvg	1329	154	111	81	84	80	74	75	102	92	149	166	157	(15)
(16)		PrecMax	1622	255	259	151	168	189	177	152	244	166	275	283	301	(16)
(17)		PrecMin	1118	11	30	24	21	11	19	20	8	16	42	104	27	(17)
(18)		PrecStd	145	64	59	40	39	40	45	32	55	36	58	54	72	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.6	14.0	17.2	23.2	23.2	25.9	27.5	21.2	18.5	14.5	12.7	(19)
(20)			MCWB	10.1	9.5	10.7	11.8	16.5	16.6	18.1	20.0	16.8	15.3	12.8	11.7	(20)
(21)		2%	DB	10.7	10.6	11.9	14.7	20.7	21.2	23.7	24.1	19.2	16.8	13.7	12.1	(21)
(22)			MCWB	9.7	9.2	9.6	10.9	15.5	15.9	17.5	19.0	15.6	14.7	12.7	11.2	(22)
(23)		5%	DB	10.2	10.0	11.0	13.1	18.7	19.1	21.9	21.6	18.2	15.9	13.1	11.4	(23)
(24)			MCWB	9.3	8.9	9.1	10.1	14.6	14.4	16.7	17.5	15.2	14.3	12.1	10.4	(24)
(25)		10%	DB	9.7	9.5	10.3	11.9	16.6	17.4	20.1	20.0	17.2	15.2	12.5	10.7	(25)
(26)			MCWB	8.8	8.4	8.7	9.4	12.9	13.8	16.2	16.8	14.8	13.7	11.4	9.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.6	11.3	13.3	17.5	17.8	19.8	20.6	17.9	16.3	13.7	12.2	(27)
(28)			MCDWB	10.9	11.0	12.5	16.7	21.9	21.4	24.2	25.6	19.7	17.5	13.9	12.5	(28)
(29)		2%	WB	10.1	9.7	10.5	11.6	15.8	16.3	18.5	19.4	16.7	15.4	13.0	11.6	(29)
(30)			MCDWB	10.4	10.2	11.5	13.5	20.0	19.9	21.7	22.9	18.2	16.2	13.4	12.0	(30)
(31)		5%	WB	9.7	9.2	9.8	10.7	14.6	15.2	17.6	18.4	16.0	14.7	12.4	10.7	(31)
(32)			MCDWB	10.1	9.8	10.6	12.3	18.2	17.9	20.3	20.9	17.2	15.6	12.9	11.3	(32)
(33)		10%	WB	9.2	8.7	9.2	10.0	13.5	14.5	16.8	17.4	15.4	14.1	11.7	9.9	(33)
(34)			MCDWB	9.7	9.4	10.0	11.3	16.1	16.6	19.2	19.5	16.8	15.1	12.4	10.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	5.5	6.7	7.6	6.7	7.2	7.2	6.9	6.0	5.7	5.0	(35)
(36)			MCDBR	4.8	4.9	5.9	7.8	10.8	10.5	10.8	9.5	8.5	5.7	4.8	4.1	(36)
(37)			MCWBR	4.8	4.2	4.4	5.1	6.6	6.4	6.1	5.7	5.7	4.5	4.2	4.3	(37)
(38)		5% WB	MCDWR	4.8	4.2	5.2	6.7	9.8	8.2	8.5	8.6	7.3	5.3	4.1	4.0	(38)
(39)			MCWBR	5.0	4.0	4.4	5.0	6.2	5.5	5.4	5.5	5.7	4.8	3.9	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.346	0.375	0.389	0.397	0.385	0.389	0.372	0.366	0.356	0.332	0.324	(40)	
(41)		taud	2.324	2.368	2.309	2.296	2.319	2.382	2.378	2.440	2.447	2.430	2.379	2.302	(41)	
(42)		Ebn at Noon	657	768	813	847	856	869	859	857	819	748	658	598	(42)	
(43)		Edh at Noon	65	81	104	117	120	114	113	101	90	76	61	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.74	1.51	2.75	4.39	5.39	5.87	5.34	4.46	3.30	1.83	0.92	0.56	(44)	
(45)		RadStd	0.07	0.14	0.24	0.31	0.34	0.49	0.45	0.30	0.24	0.15	0.07	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page